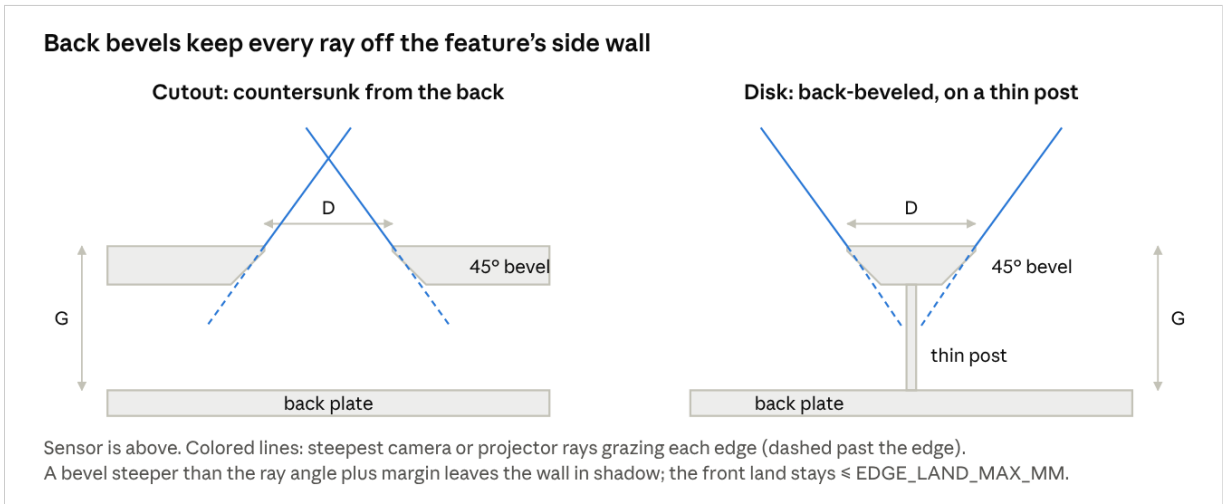


3.3 Chamfered (knife-edge) boundaries

Every boundary that defines an edge, disk, or cutout must be chamfered from the back. Then no camera or projector ray can strike the boundary's side wall, and the sensor sees only the front face and the back plate. An unchamfered wall would be seen as a third surface by one camera and not the other, which corrupts exactly the edge measurements this procedure is meant to make.

- Cutouts and the square window: countersink from the back face, so the hole widens away from the sensor.
- Disks and the raised square: bevel the back, so the part narrows away from the sensor (a frustum).
- Land: the flat land left at the front edge must be $\leq \text{EDGE_LAND_MAX_MM}$.
- Bevel angle: measured from the plate normal, it must exceed the worst-case ray angle plus a margin:



Chamfered cutout and disk, cross-section (not to scale)

The steepest ray past each edge travels through open space and never meets the beveled wall, so the sensor sees only the front face and the back plate.

$$\theta_{\text{bevel}} \geq \max_{c \in \{L, R, P\}} \max_{e \in \text{pose}} \angle \big(\mathbf{e} - \mathbf{c}, \hat{\mathbf{n}}_{\text{plate}} \big) + \theta_{\text{margin}}$$

Here c is the left camera, right camera, or projector center, e is any edge point at any pose used, and $\theta_{\text{margin}} = \text{CHAMFER_MARGIN_DEG}$. A rough worked value, assuming about a 70° horizontal FOV and a 50 mm baseline (both unconfirmed): the worst field position at Z_{MIN} gives about 32° plus a 10° margin, or 42° . A standard 45° bevel (90° included countersink) would then pass. Recompute this once the VSX3000 geometry is known. If the value exceeds 45° , use a steeper bevel or reduce $\text{FIELD_OFFSET_FRACTION}$ for the C, D, and E series.

Disk support posts. The post must sit behind the disk and be thinner than $\text{POST_DIAMETER_FRACTION_OF_D0} \times$ the expected D_0 of Section 3.2 (about 7 px, 4 mm at Z_{MIN} for $f_x \approx 688$ px), so posts of 2 mm or less. The post-only control sites confirm the post is not detected.

As-built record. Measure each feature's front-face diameter, land width, bevel angle, and plate position with an optical comparator or calibrated microscope. Record each value with its measurement uncertainty in `targets_asbuilt.csv`. All analyses use these as-built values, never the nominal ones.